

Resting-State Power and Regional Connectivity After Pediatric Mild Traumatic Brain Injury

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Background: Physiological recovery from pediatric mild traumatic brain injury (pmTBI) as a function of age remains actively debated, with the majority of studies relying on subjective symptom report rather than objective markers of brain physiology.

Purpose: To examine potential abnormalities in fractional amplitude of low-frequency fluctuations (fALFF) or regional homogeneity (ReHo) during resting-state fMRI following pmTBI.

Study Type: Prospective cohort.

Population: Consecutively recruited pmTBI ($N = 105$; 8–18 years old) and age- and sex-matched healthy controls (HC; $N = 113$).

Field Strength/Sequence: 3T multiecho gradient T_1 -weighted and single-shot gradient-echo echo-planar imaging.

Assessment: All pmTBI participants were assessed 1 week and 4 months postinjury (HC assessed at equivalent timepoints after the first visit). Comprehensive demographic, clinical, and cognitive batteries were performed in addition to primary investigation of fALFF and ReHo. All pmTBI were classified as “persistent” or “recovered” based on both assessment periods.

Statistical Tests: Chi-square, nonparametric, and generalized linear models for demographic data. Generalized estimating equations for clinical and cognitive data. Voxelwise general linear models (AFNI’s 3dMVM) for fALFF and ReHo assessment.

Results: Evidence of recovery was observed for some, but not all, clinical and cognitive measures at 4 months postinjury. fALFF was increased in the left striatum for pmTBI relative to HC both at 1 week and 4 months postinjury; whereas no significant group differences ($P > 0.001$) were observed for ReHo. Age-at-injury did not moderate either resting-state metric across groups. In contrast to analyses of pmTBI as a whole, there were no significant ($P > 0.001$) differences in either fALFF or ReHo in patients with persistent postconcussive symptoms compared to recovered patients and controls at 4 months postinjury.

Data Conclusions: Our findings suggest prolonged clinical recovery and alterations in the relative amplitude of resting-state fluctuations up to 4 months postinjury, but no clear relationship with age-at-injury or subjective symptom report.

Level of Evidence: 1

Technical Efficacy: 2

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THE OCCURRENCE OF TRAUMATIC BRAIN INJURY (TBI) represents a prevalent societal health concern among neurological disorders, affecting ~1.7 million people annually.¹ Of those, a considerable proportion of injuries occur in children and are considered to be mild in nature (pediatric mild TBI; pmTBI used synonymously here with concussion; 2). The exact number of children who continue to exhibit postconcussive symptoms (PCS) 1 month or more postinjury (henceforth identified as persistent-PCS) remains actively debated.² Some estimates suggest 30–35% of children do not recover by 4 months postinjury,³ whereas others indicate a lower (10–15%) estimate⁴ when accounting for the nonspecific nature of symptom reporting.⁵

Recently, it has been suggested that resting-state connectivity⁶ may be more sensitive than self-report measures for not only detecting the acute to subacute effects of injury, but also for identifying patients who may remain symptomatic.^{7,8} Previous resting-state analyses have primarily utilized a priori selected regions that are vulnerable to mTBI, with findings of increased connectivity within the posterior cingulate cortex (PCC)/precuneus node of the default mode network (DMN) in pmTBI relative to controls.^{9,10}

While the theoretical basis for the selection of these regions is well established,¹¹ whole-brain voxelwise approaches like regional homogeneity (ReHo¹²) and fractional amplitude of low-frequency fluctuations (fALFF¹³) also have several benefits. Foremost, they do not require multiple corrections for familywise error if looking across multiple regions and may better account for the heterogeneous nature of mTBI at the local level. Although previous studies have examined ReHo and fALFF alterations in both sport and emergency department (ED) samples of adult mTBI patients,^{8,14,15} there is a lack of data using these parameters to characterize the effects of mTBI in children.¹⁶

The first aim of our study was to examine whether ReHo or fALFF were affected and subsequently recovered in a large cohort of pmTBI at subacute (SA; ~1 week) and early chronic (EC; ~4 months) postinjury periods relative to age- and sex-matched healthy controls (HCs). The second aim was to examine whether these metrics varied between recovered pmTBI and those presenting with persistent-PCS at 4 months postinjury. We hypothesized that ReHo and fALFF abnormalities would be primarily localized to midline hub regions of the DMN (rostral anterior cingulate cortex [rACC] and posterior cingulate cortex [PCC]) and subcortical structures,^{8–10} and would be associated with persistent symptomatology. Secondary analyses also examined whether older age-at-injury represented a potential moderating effect for ReHo and fALFF metrics, as has been observed in the clinical literature.³

Materials and Methods

The study was approved by the University of New Mexico Institutional Review Board and participants provided informed consent

(assent for participants 8–11 years old) per University of New Mexico School of Medicine guidelines.

Participants

Participants with pmTBI were consecutively recruited from local emergency department and urgent care settings between April 2016 and July 2019 in this prospective cohort study. Data collection was attempted at two visits during subacute (SA: ~1–11 days postinjury) and early chronic (EC: ~4 months postinjury) injury phases in this ongoing study. Inclusion criteria were based on the American Congress of Rehabilitation Medicine and the Zurich Concussion in Sport Group guidelines. Specifically, pmTBI were enrolled if they experienced a closed head injury with Glasgow Coma Score ≥ 13 , loss of consciousness (if present) limited to 30 minutes, posttraumatic amnesia (limited to 24 hours), alteration in mental status at the time of injury, or at least two new PCS. Age- and sex-matched HC were recruited and assessed at corresponding timepoints to control for neurodevelopmental confounds and/or repeat assessment.

Exclusion criteria for all participants included history of 1) previous head injury with greater than 30-minute loss of consciousness; 2) a neurological diagnosis; 3) history of any psychiatric disorders other than adjustment disorder; 4) developmental disorder (autism spectrum disorder or intellectual disability); 5) history of substance abuse/dependence; 6) contraindications for magnetic resonance imaging (MRI) including pregnancy; or 7) non-English-speaking. Additional exclusion criteria for HC included a diagnosis of Attention-Deficit Hyperactivity Disorder or a learning disability. Patients were also excluded if general anesthesia was administered immediately following injury or if the injury affected the dominant hand. Urine-based drug screens (see Appendix S1 in the Supplemental Material) were conducted for all participants at each visit, and a positive test resulted in exclusion.

Clinical Measures

A Common Data Elements-focused battery of clinical and neuropsychological measures (see Appendix S1 in the Supplemental Material for full battery) was administered at both the SA and EC periods. Retrospective ratings (ie, estimation for 1 month ago) of symptoms were also acquired at the SA visit (see Table S2 in the Supplemental Material for retrospective clinical ratings across HC and pmTBI groups). Primary clinical measures included the Post-Concussion Symptom Inventory (PCSI; self and parental report),^{17,18} Conflict and Behavioral Questionnaire (CBQ),¹⁹ and the Pediatric Quality of Life Inventory (PedsQL; Generic Core).²⁰ Secondary clinical measures consisted of the Patient Reported Outcomes Measurement Information System (PROMIS; sleep,²¹ anxiety and depression²² scales), a self-report pain rating (0–10 Likert scale),²³ Headache Impact Test,²⁴ the Strengths and Difficulties Questionnaire,²⁵ and self-reported Tanner stage of development.²⁶ Parental reports of psychopathology were measured with the Brief Symptom Inventory.²⁷ Overall outcomes were assessed with the Glasgow Outcome Scale Extended (Pediatric Revision).²⁸

Cognitive estimates of premorbid abilities and effort were obtained along with neuropsychological tests selected from specific subtests of the Delis-Kaplan Executive Function System²⁹ and age appropriate versions of Wechsler scales (see Appendix S1 in the Supplemental Material).^{30,31} Raw scores were age-corrected and

aggregated into composites for primary domains of attention and processing speed, along with secondary domains of working memory and executive function. Performances on the Wide Range Achievement Test 4 (WRAT4)³² and Test of Memory Malingering (TOMMe10)³³ were also included as secondary cognitive measures.

Classification of persistent-PCS from recovered patients at the 4-month follow-up visit (EC) was determined based on self-reported PCSI data.⁴ Briefly (see Appendix S1 in the Supplemental Material), individual HC and pmTBI PCSI data at the EC visit were summed, base 10 log-transformed to correct for nonnormality, and further corrected relative to HC for distributional bias in nonindependent Z-scores ($Z_{\text{HC[persistent-PCS]}} > 1.64$). Those pmTBI above the threshold were identified as having persistent-PCS, while those below the threshold were classified as recovered.

MRI Parameters

All participants were scanned with a 3T Trio system (Siemens; Erlangen, Germany) with a 32-channel head coil. Foam padding was used to minimize head motion. A high-resolution five-echo magnetization prepared rapid acquisition gradient echo (MPRAGE) T₁-weighted (repetition time [TR] = 2530 msec; echo time [TE] = 1.64, 3.50, 5.36, 7.22, 9.08 msec; inversion time [TI] = 1200 msec; flip angle = 7°; number of excitations [NEX] = 1; slice thickness = 1 mm; field of view [FOV] = 256 mm; matrix size = 256 × 256; isotropic voxels = 1 mm³) sequence was collected in addition to a T₂-weighted sequence (TR = 15500 msec; TE = 77 msec; flip angle = 155°; NEX = 1; slice thickness = 1.5 mm; FOV = 220 mm; matrix size = 192 × 192; voxel size = 1.15 × 1.15 × 1.5 mm³). Resting-state data were acquired over a 5-minute period using a single-shot, gradient-echo echoplanar pulse sequence (TR = 460 msec; TE = 29 msec; flip angle = 44°; multiband acceleration factor = 8; NEX = 1; slice thickness = 3 mm; FOV = 248 mm; matrix size = 82 × 82) with 56 interleaved 3-mm slices acquired for whole-brain coverage (voxel size: 3.02 × 3.02 × 3.00 mm³). A total of 647 images were analyzed following the removal of initial images to account for T₁ equilibrium effects. Due to increased gray-white contrast, a reference image with multiband acceleration factor set to one (ie, no acceleration) was acquired to facilitate registration with the native T₁-weighted anatomical image. Finally, two spin-echo field mapping sequences (TR = 7220 msec; TE = 73 msec; flip angle = 90°; refocus flip angle = 180°; slice thickness = 3 mm; FOV = 248 mm; matrix size = 82 × 82; 56 interleaved slices; 3.02 × 3.02 × 3.00 mm³ voxels) with reversed phase encoding directions (A→P; P→A) were collected to account for susceptibility distortions in the gradient echo data. Finally, a subset of pmTBI participants ($N = 45$) received computed tomography (CT) scans as part of routine care. These individuals were assessed for corresponding intracranial trauma across CT and structural MRI scans.

Resting-State Data Processing

All image processing steps were implemented using AFNI programs unless otherwise specified (see Appendix S1 in the Supplemental Material). Briefly, data were first despiked, followed by temporal interpolation to the first slice and spatial registration in 2D and 3D space to the single-band reference image. Next, images were corrected for susceptibility field distortions and then registered to the

subject-specific T₁, which, in turn, was registered nonlinearly to a stereotaxic template. Six motion parameters, their derivatives, average white matter, and cerebral spinal fluid (CSF) signals were regressed as nuisance variables from the spatially normalized maps.

AFNI's 3dRSFC program^{13,34} was used to generate fALFF and bandpass-filtered data (0.01–0.10 Hz) based on the following steps: 1) transformation of each voxel time series to a frequency domain (0–0.25 Hz) without bandpass filtering and taking the square root of each frequency of the power spectrum; and 2) dividing the sum amplitude of the low-frequency domain (0.01–0.10 Hz) by the full frequency domain (0–0.25 Hz). AFNI's 3dREHO program^{12,34} calculated Kendall's concordance coefficient (KCC) across a neighborhood of 27 native voxels on the bandpass-filtered data to produce the ReHo images. fALFF and ReHo images were smoothed with a 6-mm full-width at half-maximum (FWHM) kernel and converted to Z-scores¹³ before being transformed nonlinearly into standard stereotaxic Talairach space.

Statistical Analysis

Demographic data were evaluated using the chi-square (sex and previous history of concussions) and Mann–Whitney U (age-at-injury) tests, while primary and secondary clinical and cognitive data were assessed with either generalized linear models (GLM; Group only) or generalized estimating equations (GEE; Group × Visit), using gamma or negative binomial distributions, as appropriate. The effects of typical recovery on fALFF and ReHo whole-brain metrics were evaluated using a 2 × 3 GLM (Group [HC vs. pmTBI] × Time [SA vs. EC]), whereas specific effects associated with persistent-PCS were evaluated with a one-factor GLM (HC vs. recovered-pmTBI vs. persistent-PCS) from the EC visit. All analyses were conducted using AFNI's 3dMVM module with mean framework displacement (FD), which is the average of frame-by-frame translational and rotational head motion across all three imaging runs, and age-at-injury entered as covariates. Prior to analyses, all participants reporting greater than three times the interquartile range above the median on mean FD relative to their cohort were excluded from analyses. All demographic and behavioral/clinical/cognitive analyses were assessed for significance at $P \leq 0.05$, while all whole-brain imaging results were corrected for familywise error using parametric ($P \leq 0.001$) and minimum volume thresholds (972 μL) based on results from 10,000 Monte Carlo simulations and spherical autocorrelation estimates.³⁵ Those clusters surviving significance and minimum volume thresholds from Group or Group × Visit 3dMVM models were subjected to additional exploratory analyses using linear mixed effects models with primary clinical (Bonferroni correction at 0.05/3 = 0.0167) and cognitive measures (Bonferroni correction at 0.05/2 = 0.025) within pmTBI to determine associations with fALFF/ReHo.

Finally, subject-specific abnormalities in fALFF and ReHo were assessed using previously described methods.³⁶ Voxelwise data from pmTBI and HC at all visits were transformed to Z-scores with the mean and standard deviation of the HC group at the SA visit serving as the reference group. Z-scores were distribution-corrected (DisCo-Z) based on the number of participants in the reference group to control for known biases following transformation. The number and volume of clusters with abnormally increased or decreased fALFF or ReHo were calculated, with abnormal clusters

defined as clusters with a minimum size of 275 μL (approximate volume of 27.43 native space voxels) with $Z < -2$ (negative cluster) or $Z > +2$ (positive cluster³⁷) and assessed with 2×3 GEE (Group: HC vs. pmTBI $\times$ Time: SA vs. EC) models with significance determined at $P \leq 0.05$.

Results

Participants

Nine individuals (four pmTBI and five HC) withdrew from the study during the course of their first visit, while another four HC presented with additional findings or issues (eg, neurological findings on structural MRI), excluding them from further participation in the study. No attempt was made to bring these participants back for a second visit, resulting in a potential sample of 181 pmTBI and 162 HC for return appointments. For the purposes of the current study, 158/181 pmTBI and 135/162 HC were eligible for the 4-month follow-up visit (ie, larger parent study is ongoing).

Of these participants, 129 pmTBI (81.6% retention) and 127 HC (94.1% retention) returned for their 4-month visit. However, several additional participants (nine pmTBI, three HC) did not complete the MRI portion of the experiment for various reasons (new braces, pregnancy, etc.), were missing key clinical data (five pmTBI, three HC), or tested positive for tetrahydrocannabinol (four pmTBI). Finally, six pmTBI and eight HC were excluded as motion outliers on the resting-state scan at either visit. Thus, the final sample with usable imaging data for both visits included 105 pmTBI (47 females; age 14.04 ± 2.7 ; 7.3 ± 2.2 initial days postinjury; days between visits 124.3 ± 14.8) and 113 HC (51 females; age 14.54 ± 2.5 ; days between visits 124.4 ± 15.1). See Table S1 in the Supplemental Material for demographic information across groups at each eligibility phase of the study.

Typical Recovery

DEMOGRAPHICS AND CLINICAL MEASURES. There were no differences between HC and pmTBI groups on any demographic variables (ie, sex, age, and handedness). There was a main effect of Group (pmTBI > HC) from the GLM model of previous history of mild head injuries (Wald- $\chi^2 = 7.41$, $P = 0.007$). See Table 1 for full demographic and clinical/cognitive performance metrics and exact P -values.

The analysis for self-report PCS with GEE models, controlling for retrospective ratings, indicated a significant Group $\times$ Visit interaction (Wald- $\chi^2 = 13.30$, $P \leq 0.001$) with follow-up testing indicating that pmTBI reported significantly more symptoms at the SA ($P \leq 0.001$) and EC ($P = 0.002$) visits relative to HC, but with a larger magnitude of effect at the SA visit (Fig. 1a). GEE models of self-reported behavioral changes (Group $\times$ Visit: $P = 0.087$; Group: $P = 0.778$), as well as quality of life (Group: $P = 0.168$; assessed only at the EC visit), were not significant (Fig. 1b).

Interactions for secondary clinical measures (Fig. S1A in the Supplemental Material) of anxiety ($P = 0.014$), sleep disturbances ($P = 0.007$), headaches ($P = 0.001$), and pain ($P \leq 0.001$) were all significant (see full results in Appendix S1 in the Supplemental Material), with follow-up analyses indicating group differences (pmTBI > HC) at SA but not EC (pmTBI $\approx$ HC) visits (see Table 1 for exact P -values of GEE pairwise follow-up tests). In contrast, an overall main effect of Group was observed for depression ($P = 0.006$).

For primary cognitive outcomes (Fig. 1c), there was a Group $\times$ Visit interaction for attention (Wald- $\chi^2 = 4.66$, $P = 0.031$), with follow-up testing indicating group differences at SA (pmTBI < HC; $P = 0.003$) but not EC (pmTBI $\approx$ HC; $P = 0.087$) visits. Additionally, there was a Group main effect for lower processing speed in pmTBI relative to HC (Wald- $\chi^2 = 4.27$, $P = 0.039$). For secondary cognitive measures (Fig. S1B in the Supplemental Material), there was a Group $\times$ Visit interaction for TOMMe10 (Wald- $\chi^2 = 4.81$, $P = 0.028$), with pmTBI reporting lower scores relative to HC at SA ($P = 0.004$) but not EC ($P = 0.819$) assessments. Group effects were observed for working memory ($P = 0.016$) and executive function ($P = 0.003$), with pmTBI exhibiting overall lower performance relative to HC (see Table 1 for exact P -values of GEE pairwise follow-up tests). Finally, a Group effect was observed for the WRAT4 (Wald- $\chi^2 = 9.24$, $P = 0.002$; assessed only at SA visit), with pmTBI reporting lower values than HC.

REHO AND FALFF RESULTS. A total of 45 pmTBI received a CT scan as part of routine care, with 4/45 pmTBI diagnosed with a positive intracranial finding on CT. Three additional patients were identified as having trauma-related findings on structural MRI scans (Table S3 in the Supplemental Material). Initial analysis indicated an overall group (pmTBI > HC) difference in FD (Wald- $\chi^2 = 7.700$, $P = 0.006$). FD was also inversely associated with age ($\rho = -0.468$; $P \leq 0.001$). FD was therefore entered as a nuisance covariate for all imaging analyses.

Results from the 2×2 (Group [HC vs. pmTBI] $\times$ Visit [SA vs. EC]) GLM showed a significant increase in fALFF ($P \leq 0.001$) for pmTBI relative to HC across visits within the left striatum (caudate and putamen; $2145 \mu\text{L}$; Fig. 2). There were no additional Group interactions with Age and/or Visit (all $P > 0.001$). Subject-specific analyses for fALFF did not demonstrate any Group effects or interactions for positive/negative cluster count or total volume (all $P > 0.05$). Exploratory linear mixed effects analyses were performed to determine if fALFF was associated with primary clinical (Bonferroni correction at $0.05/3 = 0.01667$) or cognitive measures (Bonferroni correction at $0.05/2 = 0.025$) within pmTBI. However, the results were null for all clinical (PCS: $P = 0.843$; behavioral changes: $P = 0.746$; quality of

TABLE 1. Clinical and Neuropsychological Data (HC vs. pmTBI) Across Visits

Metric	Outcome	SA HC	SA pmTBI	SA- <i>p</i>	EC HC	EC pmTBI	EC- <i>p</i>
<i>Demographics</i>							
Age (years)	D	15 (13–16)	14 (12–16)	0.174	—	—	—
Sex (% Female)	D	45.13%	44.76%	0.956	—	—	—
Prev Hx of mTBI	D	8.8%	21.9%	0.007	—	—	—
<i>Symptom Measures</i>							
PCSI	P	2 (0–8)	11 (3–35)	≤0.001	3 (1–8)	4 (0–17)	0.002
PROMIS Sleep	S	14 (10–17)	17 (13–23)	≤0.001	15 (12–18)	18 (12–23)	0.116
PROMIS Anxiety	S	1 (0–4)	3 (0–8)	≤0.001	1 (0–4)	2 (0–6)	0.158
PROMIS Depression	S	0 (0–3)	2 (0–8)	0.004 ^a	1 (0–3)	1 (0–5.5)	0.119 ^a
Pain Scale	S	0 (0–1)	3 (1–5)	≤0.001	0 (0–1)	0 (0–2)	0.135
HIT-6	S	40 (36–47)	50 (42–58)	≤0.001	42 (38–47)	44 (40–53.5)	0.237
<i>Behavioral & Outcome Measures</i>							
CBQ	P	1 (0–2)	1 (0–2)	0.141	1 (0–2)	1 (0–3)	0.404
PedsQL	P	—	—	—	90.22 (82.07–95.65)	84.78 (76.09–93.48)	0.168
<i>Cognitive Measures</i>							
TOMMe10	S	9.86 ± 0.42	9.57 ± 1.21	0.004	9.79 ± 0.47	9.77 ± 0.58	0.819
WRAT4	S	54.0 (48.7–62.7)	49.3 (44.7–55.3)	0.002	—	—	—
PS	P	50.37 ± 7.94	47.82 ± 7.3	0.014 ^b	52.66 ± 8.46	51.08 ± 8.2	0.160 ^b
AT	P	51.1 (45.53–55.53)	47.78 (42.23–53.33)	0.003	52.2 (47.77–56.67)	50 (45.53–54.43)	0.087
WM	S	50 (43.3–56.7)	46.7 (41.65–51.65)	0.010 ^c	50 (43.3–56.7)	48.76 ± 9.59	0.069 ^c
EF	S	49.71 ± 6.68	47.53 (44.18–51.66)	0.006 ^d	53.35 (50.83–56.68)	51.65 (45.83–55.03)	0.004 ^d

AT: Attention; CBQ: Children's Behavior Questionnaire; D: demographic; EC: Early chronic; EF: Executive Functioning; HC: healthy control; HIT-6: Headache Impact Test; Hx: history; P: primary; PCSI: Post-Concussion Symptom Inventory; PedsQL: Pediatric Quality of Life Inventory; pmTBI: pediatric mild traumatic brain injury; PROMIS: Patient-Reported Outcomes Measurement Information System; PS: Processing Speed; S: secondary; SA: Subacute; TOMMe10: Test of Memory Malingering- Errors on First Ten Items; WM: Working Memory; WRAT4: Wide Range Achievement Test Fourth Edition. Data are either formatted as mean ± standard deviation or median (interquartile range).

^aOmnibus Group effect significant at $P = 0.006$.

^bOmnibus Group effect significant at $P = 0.039$.

^cOmnibus Group effect significant at $P = 0.016$.

^dOmnibus Group effect significant at $P = 0.003$.

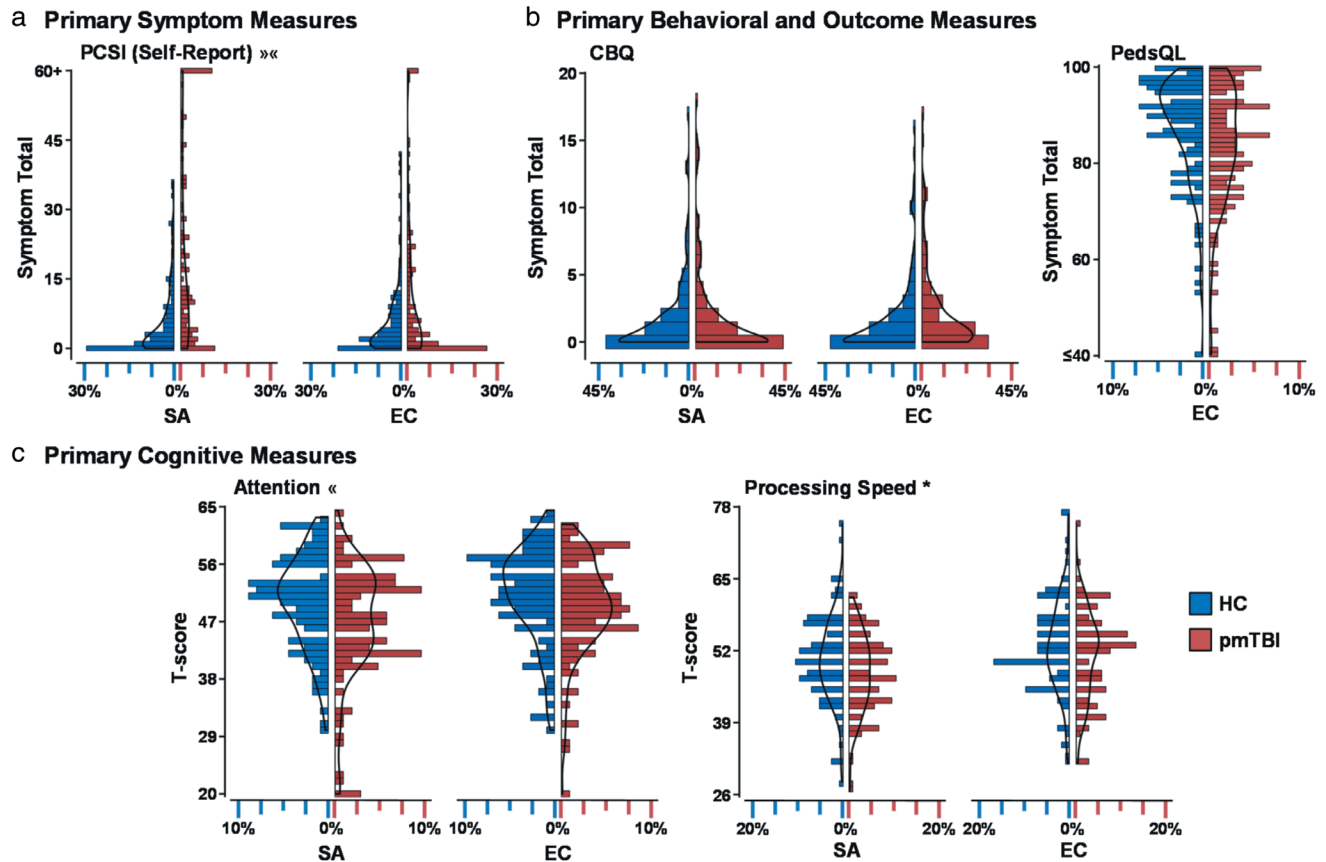


FIGURE 1: Distributions of primary clinical and cognitive measures. Histograms (bin = 1) present the percentage (x-axis) of healthy controls (HC; blue) and pediatric mild traumatic brain injury patients (pmTBI; red) with symptom total scores (a), behavior and outcome scores (b) and cognitive performance (c) t-scores (y-axis) from primary measures across subacute (SA) and early chronic (EC) assessments. Density plots (solid black line) are overlaid on all histograms. The x-axis is scaled to approximate the maximum percentage value among the four distributions (or two distributions for PedsQL) for each symptom inventory and cognitive performance t-score across both visits. The Post-Concussion Symptom Inventory (PCSI), Conflict and Behavior Questionnaire (CBQ), and Pediatric Quality of Life measure (PedsQL) in (a) and (b) were all self-report. Panel (c) presents composite t-scores for cognitive performance on assessments of attention and processing speed, respectively. Significant Group effects (*) and Group $\times$ Time interactions (« = group differences at SA only; » = differences in magnitude across SA and EC periods) are denoted.

life: $P = 0.475$) and cognitive (attention: $P = 0.745$; processing speed: $P = 0.444$) terms within the models.

There were no significant Group effects, Group $\times$ Visit or Group $\times$ Age interactions for ReHo. For subject-specific analyses, there were no significant Group effects or interactions regarding both positive and negative cluster count and volume for either ReHo or fALFF (all $P > 0.05$).

Although the primary interest of Age and Visit were in their association/interaction with the presence of pmTBI, Age- and Visit-only effects were observed across fALFF and ReHo maps. For Age, expansive effects were observed across bilateral cortical, subcortical, and white matter in the cerebrum, and primarily gray matter in the cerebellum (Fig. 3). Generally, fALFF and ReHo increased with Age in frontotemporal and subcortical gray and white matter regions and decreased with Age in the posterior cerebral and cerebellar regions across fALFF and ReHo, in addition to an inverse age association for ReHo and the superior temporal gyrus.

Concerning overall factor effects of Visit, there was a single cluster of decreased fALFF (Fig. 4a,c) within the right anterior insula / ventrolateral prefrontal cortex (SA > EC; BAs 13/45; 1530 μ L). Several regions exhibited decreased ReHo (Fig. 4b,d) at the EC relative to SA visit, including the right (BA 20/37; 1883 μ L) and left (BA 37; 1154 μ L) inferior/middle temporal gyrus, bilateral precuneus (BAs 7/19; 2181 μ L), right (1625 μ L) and left (1306 μ L) Lobule VIIb/VIIIa of the cerebellum. In contrast, increased ReHo was observed at EC versus SA visits in the bilateral thalamus (2743 μ L).

Persistent-PCS

DEMOGRAPHICS AND CLINICAL RESULTS. Based on HC data, 86 pmTBI were classified as recovered ($Z \leq 1.64$), while 19 were classified as persistent-PCS ($Z > 1.64$) at the EC visit. There were no differences between recovered and

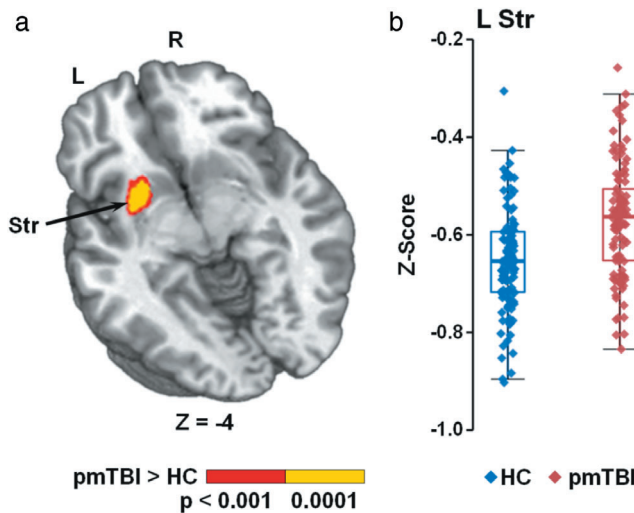


FIGURE 2: Main effect of diagnosis on fractional amplitude of low-frequency fluctuations (fALFF). (a) The position of a cluster in the left striatum (L Str) based on the Talairach atlas (Z = axial slice location; L = left; R = right). Significance level of the group effect is denoted by color intensity (red: $P \leq 0.001$; yellow: $P \leq 0.0001$). (b) Box-and-scatterplots of Z-transformed scores for the cluster comparing healthy controls (HC; $N = 113$; blue diamonds) and pediatric mild traumatic brain injury patients (pmTBI; $N = 105$; red diamonds).

persistent-PCS pmTBI across demographic variables or history of previous mTBI (see Table 2 for exact P -values).

Primary clinical and cognitive measures were compared across HC, recovered pmTBI, and persistent-PCS at the EC phase using GLM. Not surprisingly, a significant Group effect was observed for PCSI (Wald- $\chi^2 = 70.67$, $P \leq 0.001$), with follow-up testing indicating increased ratings in persistent-PCS relative to HC ($P \leq 0.001$) and recovered ($P \leq 0.001$) groups, and statistically equivalent symptomatology between HC and recovered pmTBI ($P = 0.805$). Groups did not differ in self-reported behavioral issues ($P = 0.619$) or quality of life ($P = 0.603$; see Table 2 for exact P -values of pairwise follow-up tests). For secondary clinical measures, ratings of anxiety ($P = 0.024$), sleep disturbances ($P = 0.017$), headaches ($P = 0.012$), and pain ($P = 0.003$; see full results in Appendix S1 in the Supplemental Material and Table 2 for exact P -values of pairwise follow-up group comparisons) were elevated in the persistent-PCS group (persistent-PCS > [recovered $\approx$ HC]), whereas depression ratings were not different between groups ($P = 0.135$).

There were no group differences in primary cognitive measures of attention ($P = 0.226$) or processing speed ($P = 0.371$). For secondary measures, significant Group effects were observed for executive function ($P = 0.001$; HC > [recovered $\approx$ persistent-PCS]) and on the WRAT4 ($P \leq 0.001$; assessed only at SA visit; HC > [recovered>persistent-PCS]), while no such effects were exhibited for working memory ($P = 0.061$) or the TOMMe10 ($P = 0.485$; see full results in Appendix S1 in the

Supplemental Material and Table 2 for exact P -values of pairwise follow-up group comparisons).

REHO AND FALFF RESULTS. There were no clusters exhibiting main effects for Group (HC vs. recovered vs. persistent-PCS) or interaction effects in either whole-brain ($P > 0.001$) or subject-specific ($P > 0.05$) analyses for ReHo or fALFF.

Discussion

The primary aims of our study were to determine whether metrics of regional connectivity (ReHo activity) and the relative amplitude of low-frequency resting-state fMRI fluctuations (fALFF activity) were affected by pmTBI (full [recovered and persistently symptomatic] sample) and/or were more abnormal in pmTBI with persistent symptomatology. Several clinical and cognitive measures demonstrated only partial evidence of recovery in the pmTBI group by the early chronic visit, suggestive of a longer course for full recovery. Similarly, resting-state analyses were significant for increased fALFF in the left striatum for pmTBI relative to HC at both the subacute and early chronic phase, while no significant group differences were observed for ReHo. Contrary to our a priori hypothesis, there were no ReHo or fALFF abnormalities that were specific to pmTBI with persistent symptomatology when separately examined at 4 months postinjury. Finally, we observed no evidence that the age-at-injury moderated the effects of injury or persistent symptomatology for either ReHo or fALFF metrics.

Clinical results from our study indicate that there was evidence for full recovery of anxiety, headaches, and sleep and pain disturbances for pmTBI at 4 months postinjury. In contrast, PCS symptoms showed only partial evidence of recovery, whereas depression levels remained significantly elevated in the injured group. The pmTBI group also exhibited evidence of cognitive deficits relative to HC in the domains of attention, processing speed, working memory, and executive functioning during the SA phase, with only attention exhibiting full recovery at 4 months postinjury. Notably, group differences on clinical and cognitive measures were present when accounting for noninjury factors that have previously been associated with prolonged PCS and cognitive recovery, such as premorbid psychiatric conditions and retrospective symptom ratings.³⁸ These findings are consistent with previous work demonstrating that, on average, clinical recovery for pmTBI continues beyond 1-month postinjury across multiple domains. Therefore, a multidimensional, comprehensive, and repeatable clinical evaluation is necessary to fully track pmTBI recovery.^{16,39}

Our findings indicated increased fALFF in the striatum at the subacute phase for pmTBI, which remained elevated at 4 months postinjury. Prior studies have demonstrated the sensitivity of the striatum to mTBI using a variety of

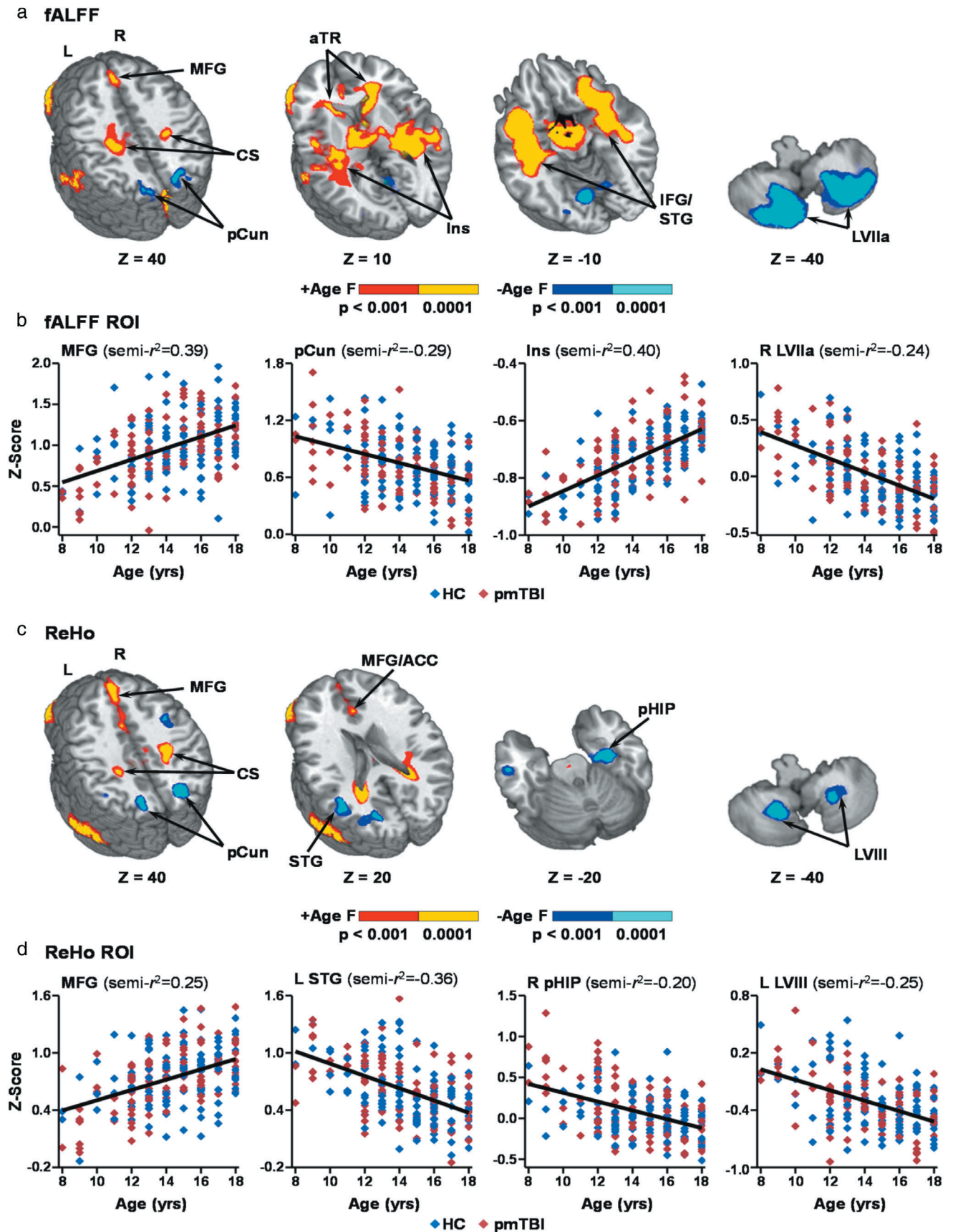


FIGURE 3: Display of clusters demonstrating the effect of age on fractional amplitude of low-frequency fluctuations (fALFF; a) and regional homogeneity (ReHo; c) activity on the Talairach atlas (Z = axial slice location; L = left; R = right). Significance level of the omnibus test and direction of age association is denoted by color intensity (positive association [red: $P \leq 0.001$; yellow: $P \leq 0.0001$]

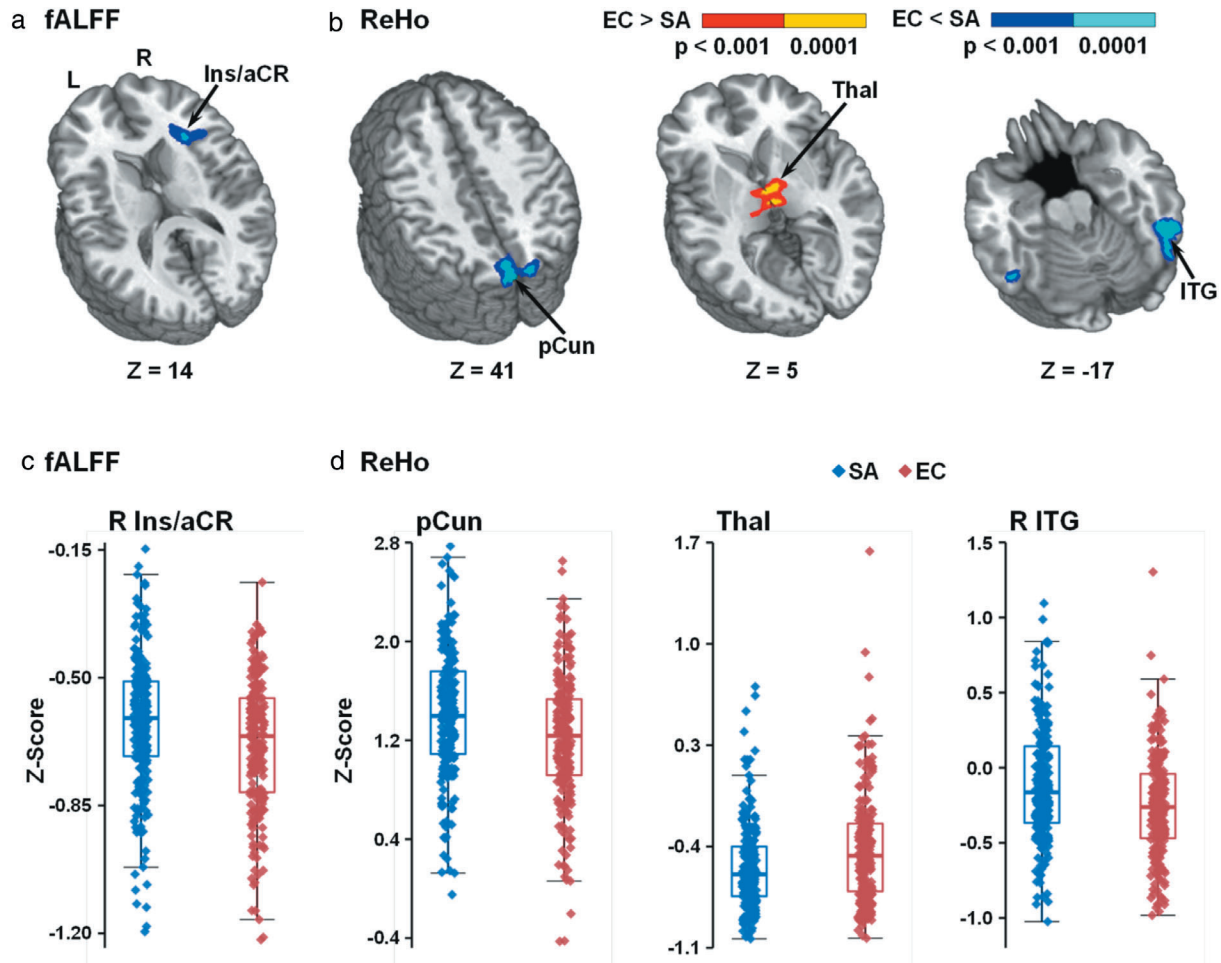


FIGURE 4: Display of clusters demonstrating a main effect of Visit on fractional amplitude of low-frequency fluctuations (fALFF; a) and regional homogeneity (ReHo; b) activity on the Talairach atlas (Z = axial slice location; L = left; R = right). Significance level and direction of the effect of Visit is denoted by color intensity (EC > SA [red: $P \leq 0.001$; yellow: $P \leq 0.0001$] and EC < SA [blue: $P \leq 0.001$; cyan: $P \leq 0.0001$]). Box-and-scatterplots present Z-transformed scores for select regions of interest (ROIs) in fALFF (c) and ReHo (d) analyses, comparing values at the subacute (SA; $N = 218$; blue diamonds) and early chronic (EC; $N = 218$; red diamonds) visits. aCR = anterior corona radiata; Ins = insula; ITG = inferior temporal gyrus; pCun = precuneus; Thal = Thalamus.

neuroimaging modalities in adult⁴⁰ and pediatric^{9,10,41} mTBI. Computational models suggest that shear stress forces from diffuse rotational injury are more likely to aggregate in these deep midline areas.⁴² Subcortical regions, including the striatum, are among the most globally and functionally connected regions of the brain.⁴³ fALFF is typically higher in cortical relative to subcortical structures,⁴⁴ and higher fALFF has been associated with increased metabolism.⁴⁵ Thus, increased fALFF in this region could reflect a potential compensatory response to damage caused by shear stress forces to the

striatum as well as affect both intra- and interhemispheric brain connectivity. However, abnormalities in the relative amplitude of resting-state blood oxygenation level-dependent (BOLD) may also result from several other factors including impaired neuronal function, impaired neural control of microvessels, direct damage to the vascular system, metabolic disruptions, or changes in cerebral blood flow.¹⁶

In contrast to fALFF findings, there were no significant differences between groups in terms of regional homogeneity. Findings from previous adult studies have been mixed, with

or negative association [blue: $P \leq 0.001$; cyan: $P \leq 0.0001$]]. Scatterplots demonstrate the association of Z-transformed scores (y-axis) and age (x-axis) for select regions of interest (ROIs) in fALFF (b) and ReHo (d) analyses across healthy controls (HC; $N = 113$; blue diamonds) and pediatric mild traumatic brain injury patients (pmTBI; $N = 105$; red diamonds), with a solid black line indicating overall fit between age and Z-transformed scores. The semi-partial correlation (semi- r^2) for age on Z-transformed scores is given, controlling for all other variables (diagnosis and mean framewise displacement). ACC = anterior cingulate cortex; aTR = anterior thalamic radiation; CS = corticospinal tract; IFG = inferior frontal gyrus; Ins = insula; LVIIa & LVIII = Lobule VIIa and VIII of the cerebellum; MFG = medial frontal gyrus; pCun = precuneus; pHIP = parahippocampus; STG = superior temporal gyrus.

TABLE 2. Clinical and Neuropsychological Data (HC vs. Recovered pmTBI vs. Persistent pmTBI) at EC Visit

Metric	Outcome	HC	Recovered	Persistent	p_{AB}	p_{AC}	p_{BC}
<i>Demographics^a</i>							
Age (years)	D	15 (13–16)	14 (12–16)	13 (13–15.5)	0.217	0.381	0.963
Sex (% Female)	D	45.13%	44.19%	47.37%	0.894	0.856	0.801
Prev Hx of mTBI	D	8.8%	20.9%	26.3%	0.015	0.042	0.759
<i>Symptom Measures</i>							
PCSI	P	3 (1–8)	3 (0–8)	39 (28.5–58.5)	0.805	≤0.001	≤0.001
PROMIS Sleep	S	15 (12–18)	17 (12–22)	24 (18–26)	0.459	0.004	0.020
PROMIS Anxiety	S	1 (0–4)	1 (0–4)	9 (3–11.5)	0.639	0.007	0.016
PROMIS Depression	S	1 (0–3)	0 (0–4)	6 (1–12)	0.259	0.054	0.311
Pain Scale	S	0 (0–1)	0 (0–1)	3 (0–4.50)	0.542	0.001	0.006
HIT-6	S	42 (38–47)	44 (40–50)	59 (52–62)	0.611	0.004	0.007
<i>Behavioral & Outcome Measures</i>							
CBQ	P	1 (0–2)	1 (0–3)	3 (1–5)	0.345	0.984	0.524
PedsQL	P	90.22 (82.07–95.65)	85.87 (77.17–93.48)	76.95 ± 15.77	0.327	0.738	0.988
<i>Cognitive Measures</i>							
TOMMe10	S	9.79 ± 0.47	9.52 ± 1.32	9.79 ± 0.54	0.703	0.282	0.232
WRAT4 ^a	S	54.0 (48.7–62.7)	49.3 (46.0–58.0)	46.0 (42.7–50.4)	0.035	≤0.001	0.002
PS	P	52.66 ± 8.46	51.01 ± 8.37	51.41 ± 7.60	0.169	0.507	0.835
AT	P	52.2 (47.77–56.67)	49.99 (45.57–54.47)	50 (45–53.88)	0.122	0.272	0.851
WM	S	50 (43.3–56.7)	50 (43.3–56.7)	43.3 (40–46.7)	0.252	0.023	0.081
EF	S	53.35 (50.83–56.68)	52.09 (45.03–56.65)	48.35 (46.27–52.51)	0.025	0.001	0.108

^aData from subacute visit.

Data are either formatted at mean ± standard deviation or median (interquartile range).

p_{AB} : HC vs. recovered pmTBI; p_{AC} : HC vs. persistent-PCS pmTBI; p_{BC} : recovered pmTBI vs. persistent-PCS pmTBI; AT: Attention; CBQ: Children's Behavior Questionnaire; D: demographic; EC: Early chronic; EF: Executive Functioning; HC: healthy control; HIT-6: Headache Impact Test; Hx: history; P: primary; PCSI: Post-Concussion Symptom Inventory; PedsQL: Pediatric Quality of Life Inventory; pmTBI: pediatric mild traumatic brain injury; PROMIS: Patient-Reported Outcomes Measurement Information System; PS: Processing Speed; S: secondary; SA: Sub-acute; TOMMe10: Test of Memory Malingering- Errors on First Ten Items; WM: Working Memory; WRAT4: Wide Range Achievement Test Fourth Edition.

decreased ReHo observed within the left insula and pre/post-central gyrus acutely¹⁴ in an ED mTBI sample, while the anterior portion of the DMN has been shown to have increased ReHo at 24 hours postinjury³⁷ and decreased ReHo at 1 month post injury⁸ in adults with sports-related concussions. Similarly, adult studies have also reported decreased

thalamic fALFF 23 days postinjury,⁴⁶ region-dependent increases and decreases acutely,⁴⁷ or null findings in fALFF in a relatively large cohort of adult athletes with acute concussion.¹⁵ These discrepancies could be due to differences in age, injury mechanism, or time since injury between the studies, all of which likely affect BOLD imaging metrics postinjury.¹⁶

Of particular relevance, inconsistencies between our findings and prior adult studies could also be explained by the dynamic and highly plastic nature of the developing brain.⁴⁸

The second aim of the study was to determine if pmTBI with persistent-PCS would exhibit more specific differences in ReHo/fALFF metrics. Not surprisingly, patients with persistent-PCS reported significant elevations in other clinical symptoms (eg, anxiety, sleep disturbances, headaches, and pain) relative to both healthy controls and the recovered pmTBI group at 4 months postinjury.^{38,39} However, persistent-PCS was not associated with any additional differences in fALFF or ReHo relative to either controls or recovered patients. Previous pmTBI studies have reported that both evoked and intrinsic midline activity in pmTBI are associated with clinical⁷ performance, with similar relationships between lower gray matter fALFF and clinical symptoms in adults.⁶

There are several potential explanations for the observed null effect on persistent-PCS. First, multiple clinical and research definitions exist for identifying patients with persistent-PCS, and these various definitions can result in a 49% spread in patient classification rates.⁴ This is likely a result of the nonspecific nature of PCS symptoms and normal fluctuations in symptom load,⁵ which complicates the identification of nonrecovered patients based on subjective complaints only.⁴⁹ When adopting more conservative criteria, the number of pmTBI classified as having persistent-PCS in the current sample ($N = 19$; 18.1%) was approximately half compared to other large-scale clinical studies using more simple clinical criteria for persistent-PCS (30–33%;³), thereby limiting statistical power. Second, a primary aim of the current study was to understand the natural course of recovery from pmTBI; therefore, participants with a past history of serious medical and psychiatric disorders were excluded. However, both of those exclusion criteria have been shown to represent potential risk factors for prolonged recovery.⁵⁰

A longstanding debate exists in the TBI literature on whether brain injury results in greater deficits (early vulnerability) or better recovery (early plasticity) as a function of age-at-injury.⁵¹ Although chronological age accounted for significant decreases/increases in both ReHo and fALFF across several cortical, subcortical, and cerebellar regions, no significant age-at-injury interactions were observed with either ReHo or fALFF data in relationship to diagnostic status. Previous studies examining the relationship between age and ReHo/fALFF in children and adolescents have been limited. One study reported that ReHo decreased with age in this developmental period, which was interpreted to represent pruning or weakening of short-range connections.⁵² Another study⁵³ reported that, although global fALFF was not associated with age in middle-aged children and adolescents (7–18 years old), fALFF within the DMN and salience network regions (insular cortex and cingulate gyrus) showed age-related decreases. Importantly, head motion was both inversely associated with age

and was greater in the pmTBI group. Although head motion parameters were included as covariates at both the subject and group level, potential effects associated with head motion status cannot be fully eliminated.⁵⁴

The majority of preclinical data examining age-at-injury is based on very young developmental periods,⁵⁵ whereas previous clinical studies have identified adolescence as a risk factor for slower clinical recovery from concussion.³ Previous diffusion and structural imaging studies showing age-related effects have been conducted in mixed severity samples⁴⁸ and suggest a complex relationship between middle childhood (greater injury coupled with increased recovery) relative to adolescents (less injury coupled with less recovery), with the latter pattern leading to more detrimental outcomes.⁴⁸ Thus, age-related injury effects on imaging metrics may be limited to structural modalities, may be more prevalent in more severely injured samples, or may require larger samples due to the dynamic nature of brain development.

Limitations

A strength of the current study was the inclusion and scanning (ie, both timepoints) of an equally powered, typically developing control cohort, reducing potential confounds in neuroimaging metrics associated with neurodevelopment. The current study also represents the largest prospective study of the effects of pmTBI on resting-state fMRI, and utilized the latest methodologies recommended for false-positive correction in imaging data.³⁵ In spite of these facts, our ability to examine the effects of persistent symptomatology was limited by power. Thus, future multisite studies are required to obtain a sufficient sample size in prospective studies to evaluate persistent-PCS. A second potential limitation concerns the duration of our resting-state scan, which was purposely shorter due to both the age of children being scanned and associated secondary effects of injury (eg, pain), both of which can result in increased motion artifact.⁵⁶ However, others suggest increased intrasession reliability and sensitivity of various connectivity metrics with longer scan lengths.⁵⁷

Conclusion

The results of our study suggest that deep subcortical structures may exhibit persistent alterations in fALFF with minimal evidence of recovery over a 4-month window, consistent with more prolonged clinical and cognitive symptoms observed in the sample. Future large-scale studies are needed to determine the extent to which alterations in subcortical resting-state fluctuations are associated with the diverse array of poor outcomes that have been identified in pmTBI.

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The authors report no competing interests.

Data Availability Statement

The data that support the findings of this study will be openly available in FITBIR at fitbir.nih.gov, reference number FITBIR-STUDY0000339 at the conclusion of this study.

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